Lucid — Complete Technical Architecture & System Methodology Manual

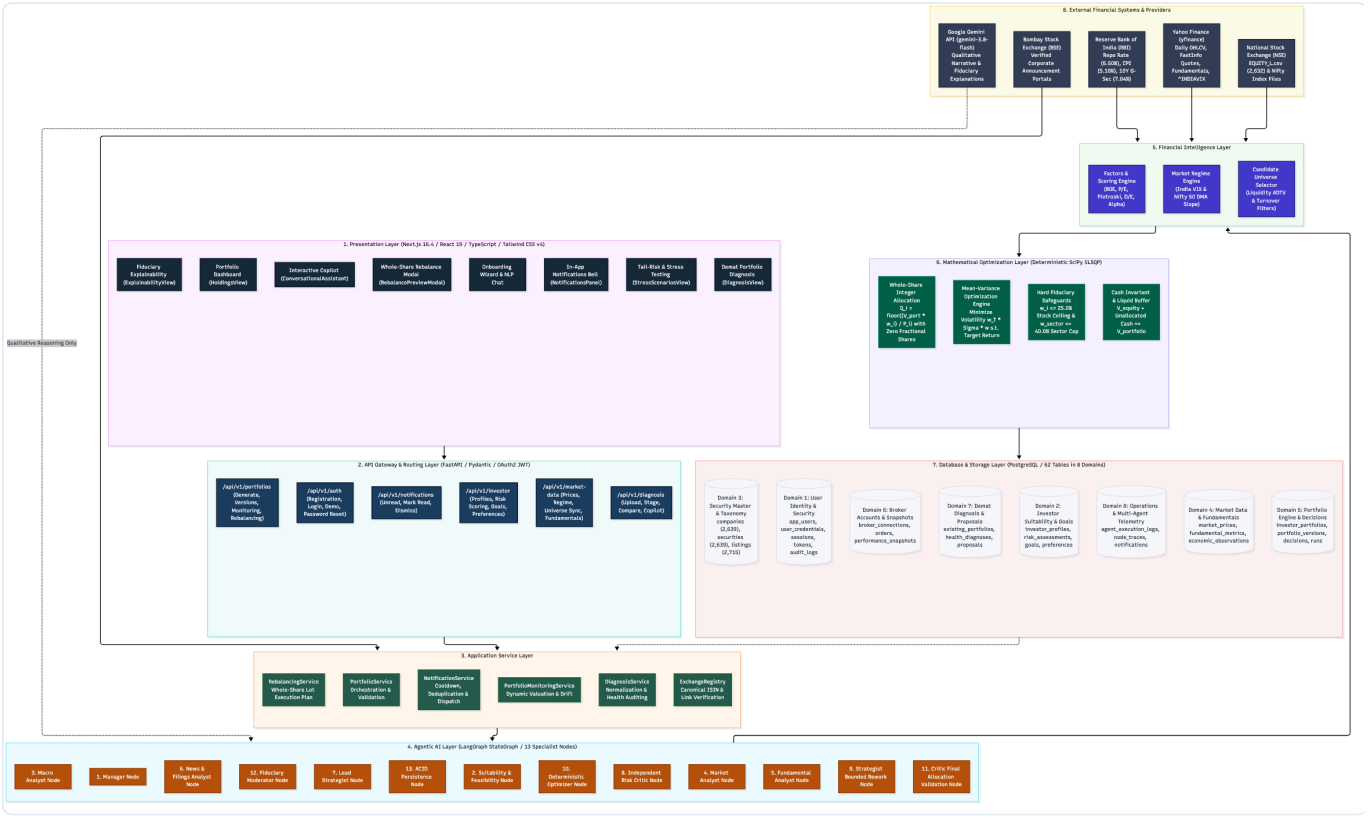
Institutional-Grade Portfolio Creation, Continuous Monitoring & Rebalancing Platform for Indian Retail Equities

Version: 2026.11-v2.0
Target Exchanges: National Stock Exchange (NSE) & Bombay Stock Exchange (BSE)
Database Persistence: PostgreSQL 16 (62 Relational Tables across 8 Domains, 2,639 Securities, 2,715 Listings)

Lucid — System Architecture Documentation

This document specifies the comprehensive 8-layer architecture of **Lucid**, an intelligent investment portfolio creation, monitoring, and rebalancing platform for Indian retail equities listed on the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE).

Architectural Blueprint (8 Layers)



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    subgraph L1 ["Layer 1: Client Presentation Layer (Next.js 16.4.2 React UI)"]
        UI_Dash["Investor Dashboard & NAV View"]:::l1
        UI_Gen["Portfolio Wizard & Risk Profiler"]:::l1
        UI_Mon["Continuous Monitoring & Drift Radar"]:::l1
        UI_Reb["Rebalance Execution & Order Preview"]:::l1
        UI_Diag["CAS Demat Statement Diagnostics"]:::l1
        UI_Chat["Copilot Chat & Fiduciary Stock-Why"]:::l1
    end

    subgraph L2 ["Layer 2: API Gateway & Middleware Layer (FastAPI 0.115)"]
        API_GW["FastAPI Core Application (backend/main.py)"]:::l2
        API_Auth["JWT Authentication & RBAC (backend/auth/)"]:::l2
        API_Rate["In-Memory Sliding-Window Rate Limiter"]:::l2
        API_Sec["Strict HTTP Security Headers (CSP, HSTS, XFO)"]:::l2
        API_Cors["CORS Policy Validation (localhost:3000)"]:::l2
        API_Routes["Modular REST Routers (v1: 9 Routers, 40+ Endpoints)"]:::l2
    end

    subgraph L3 ["Layer 3: Orchestration & State Management Layer (LangGraph)"]
        LG_Graph["StateGraph Orchestrator (backend/agent/graph.py)"]:::l3
        LG_State["PortfolioAgentState TypedDict (backend/agent/state.py)"]:::l3
        LG_Edges["Conditional Routing & Fiduciary Gate Logic"]:::l3
        LG_Trace["Full Step-by-Step Telemetry & LLM Audit Logging"]:::l3
    end

    subgraph L4 ["Layer 4: Agentic AI & Intelligence Layer (13 Specialized Nodes)"]
        A_Mgr["Manager Node (Task Decomposition)"]:::l4
        A_Suit["Suitability & Feasibility Node (SEBI Gate)"]:::l4
        A_Macro["Macro Analyst Node (RBI & Inflation)"]:::l4
        A_Mkt["Market Analyst Node (Nifty 50 Regime)"]:::l4
        A_Fund["Fundamental Analyst Node (P/E, ROE, Factors)"]:::l4
        A_News["News & Corporate Filings Analyst Node"]:::l4
        A_Strat["Lead Strategist Node (Tactical Weights)"]:::l4
        A_Critic["Independent Risk Critic Node (Mandate Audit)"]:::l4
        A_Rewk["Strategist Bounded Rework Node (Max 2 Loops)"]:::l4
        A_Mod["Fiduciary Moderator Node (Two-Level Explanations)"]:::l4
    end

    subgraph L5 ["Layer 5: Quantitative & Mathematical Engine Layer (SciPy & NumPy)"]
        M_Opt["Deterministic SLSQP Optimizer (Markowitz Quadratic Utility)"]:::l5
        M_Int["Whole-Share Integer Allocation & Cash Conservation Engine"]:::l5
        M_Drift["Asset & Sector Allocation Drift Engine (Threshold: 5%/8%)"]:::l5
        M_Cost["Statutory Indian Equity Transaction Costs Engine (STT, GST)"]:::l5
        M_Fact["Multi-Factor Scoring & Winsorized Z-Score Normalizer"]:::l5
        M_Risk["Risk Engine (Sharpe, Sortino, VaR, Max Drawdown)"]:::l5
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    subgraph L6 ["Layer 6: Operational Services & Scheduler Layer"]
        S_Cron["Background Monitoring Scheduler (300s, 09:15-15:30 IST)"]:::l6
        S_Notif["Notification Service (In-App & Push Alerts)"]:::l6
        S_Diag["Demat Diagnosis Service (pdfplumber CAS Parsing & HHI)"]:::l6
        S_Why["Stock-Why Service (Institutional Citations & Proofs)"]:::l6
        S_Reb["Rebalancing Service (Trade Preview & Invariant Checks)"]:::l6
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    subgraph L7 ["Layer 7: Data Access & Persistence Layer (PostgreSQL Master)"]
        DB_Engine["SQLAlchemy 2.0 ORM & Connection Pooling"]:::l7
        DB_Master[("PostgreSQL 16 Database (majorproject)<br/>62 Relational Tables across 8 Domains")]:::l7
        DB_Sec["Security Master (2,639 Securities, 2,715 Listings)"]:::l7
        DB_Port["Portfolio Store (Versions, Decisions, Trades)"]:::l7
        DB_Audit["Telemetry & System Audit Logs"]:::l7
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    subgraph L8 ["Layer 8: External Integrations & Market Data Layer"]
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        EXT_NSE["NSE India Archives (Official EQUITY_L Master)"]:::l8
        EXT_BSE["BSE India Corporate Directory (ISIN Cross-Reference)"]:::l8
        EXT_YF["Yahoo Finance API (yfinance Live Quotes & History)"]:::l8
        EXT_RBI["Reserve Bank of India Bulletins (Policy Repo Rate)"]:::l8
        EXT_News["NewsAPI & RSS Financial News Feeds"]:::l8
        EXT_CAS["CDSL / NSDL Consolidated Account Statement (CAS PDF)"]:::l8
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    L1 <--> L2
    L2 <--> L3
    L3 <--> L4
    L4 <--> L5
    L3 <--> L5
    L2 <--> L6
    L6 <--> L5
    L4 <--> L7
    L5 <--> L7
    L6 <--> L7
    L6 <--> L8
    L7 <--> L8
```

Detailed Architectural Breakdown by Layer

Layer 1: Client Presentation Layer

- **Framework**: Next.js 16.4.2 using React 19.3.0 and TypeScript 5.
- **Styling**: Tailwind CSS v4 with responsive layouts and accessible color palettes.
- **Key Views**:
  - `Investor Dashboard`: High-level asset allocation breakdown, performance tracking against NIFTY 50, and portfolio health indicators.
  - `Portfolio Wizard`: Step-by-step psychometric questionnaire determining risk capacity, time horizon, financial goals, and initial investment capital.
  - `Drift Radar & Live Monitoring`: Real-time visualization of individual asset and sector drift against configured policy boundaries.
  - `Rebalance Execution Preview`: Interactive breakdown of proposed integer-lot BUY, SELL, and HOLD orders, complete with statutory transaction fee estimates.
  - `CAS Demat Diagnostic View`: Drag-and-drop PDF statement uploader showing portfolio overlap, Herfindahl-Hirschman Index (HHI) concentration metrics, and stock-specific health flags.
  - `Fiduciary Stock-Why Modal`: Interactive dialogue explaining why any given security was selected or trimmed, toggleable between Plain English and Institutional Quantitative evidence.

Layer 2: API Gateway & Middleware Layer

- **Framework**: FastAPI 0.115 running on Python 3.14 with Uvicorn ASGI server.
- **Security Middleware**:
  - `InMemoryRateLimiterMiddleware`: Sliding-window rate limiter preventing brute-force authentication attacks and protecting expensive LangGraph agent invocations.
  - `Security Headers Middleware`: Enforces `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `X-XSS-Protection: 1; mode=block`, and production `HSTS`.
  - `CORS Middleware`: Whitelists trusted frontend origins (`http://localhost:3000`).
- **REST Endpoints**: 9 domain-specific routers mounted under `/api/v1` (Authentication, Profile, Goals, Risk Assessments, Market Data, Portfolios, Diagnosis, Links, Notifications).

Layer 3: Orchestration & State Management Layer

- **Orchestration Framework**: LangGraph StateGraph (`backend/agent/graph.py`).
- **State Definition**: Strongly typed `PortfolioAgentState` (`backend/agent/state.py`) capturing user profile data, research findings, agent reasoning, candidate selections, optimizer weights, critic verdicts, and audit logs.
- **Fiduciary Execution Invariants**:

- Independent Risk Critic must strictly approve before optimization begins.
- Critic Final Validation must verify that post-optimizer integer share sizing satisfies all mandate caps.
- Malformed or incomplete agent outputs trigger explicit failsafe persistence (stop-and-report).

Layer 4: Agentic AI & Intelligence Layer

- **Model Integration**: Google Gemini 1.5 Flash via `langchain-google-genai` (with Ollama fallback).
- **Specialized Roles**:
  - `Manager Node`: Decomposes investment goals, initializes graph state, and coordinates downstream specialists.
  - `Suitability & Feasibility Node`: Rule-based gatekeeper validating SEBI investor risk capacity and capital lot feasibility.
  - `Macro Analyst Node`: Assesses macro conditions (RBI repo rate 6.50%, CPI inflation 5.1%, yield curve).
  - `Market Analyst Node`: Evaluates Nifty 50 trend, 50-day and 200-day DMA breadth, and market volatility regime.
  - `Fundamental Analyst Node`: Analyzes balance sheets, P/E, P/B, ROE, ROCE, and debt-to-equity ratios.
  - `News Analyst Node`: Scans regulatory disclosures, earnings releases, and material news sentiment.
  - `Lead Strategist Node`: Synthesizes research into candidate weights and sector allocations.
  - `Independent Risk Critic Node`: Adversarially audits allocations against single-stock (10%) and sector (30%) concentration caps.
  - `Strategist Bounded Rework Node`: Enables up to 2 iterative reworks if the Critic identifies violations.
  - `Fiduciary Moderator Node`: Produces dual-level explanations (Executive Retail Summary + Institutional Factor Proofs).

Layer 5: Quantitative & Mathematical Engine Layer

- **Computational Core**: SciPy 1.15 and NumPy 2.2 executing entirely in Python without LLM hallucination risk.
- **Deterministic Optimizer**: Solves Markowitz Mean-Variance Quadratic Utility using Sequential Least Squares Programming (SLSQP).
- **Integer Lot Rounding**: Translates decimal target weights into executable integer share quantities using floor division, enforcing the strict cash conservation invariant: $$\sum_{i=1}^n (ext{Shares}_i imes ext{Price}_i) + ext{Unallocated Cash} = ext{Total Capital}$$
- **Factor Scoring**: Winsorizes outlier values at the 5th and 95th percentiles and calculates sector-relative Z-scores clamped to $[-3.0, +3.0]$.
- **Indian Statutory Costs Engine**: Accurately computes Securities Transaction Tax (STT), Exchange Turnover Fees, SEBI Turnover Fees, GST (18%), Stamp Duty, and brokerage charges.

Layer 6: Operational Services & Scheduler Layer

- **Continuous Monitoring Scheduler**: Background thread executing every 300 seconds (5 minutes) during Indian market hours (09:15 – 15:30 IST) to fetch live market quotes, revalue portfolios, and detect drift breaches.
- **Notification Service**: Dispatches transactional alerts when drift thresholds (5% single-stock or 8% sector) are breached.
- **Demat Diagnosis Service**: Employs `pdfplumber` and regular expressions to parse CDSL and NSDL CAS statements, reconciling ISINs against the Security Master.
- **Rebalancing Service**: Calculates drift turnover, synthesizes whole-share BUY/SELL/HOLD orders, and generates executable trade previews.

Layer 7: Data Access & Persistence Layer

- **Database**: PostgreSQL 16 (`majorproject`), the sole relational source of truth.
- **ORM**: SQLAlchemy 2.0 with connection pooling and request-scoped sessions.
- **Database Schema**: 62 relational tables organized into 8 functional domains:
  1. Identity & Security (7 tables)
  2. Investor Suitability & Goals (5 tables)
  3. Security Master & Companies (9 tables)
  4. Market Data, Prices & Fundamentals (11 tables)
  5. Portfolio Engine & Optimization (13 tables)
  6. Broker Integrations & Orders (6 tables)

7. Demat Diagnosis & Rebalancing (4 tables)
8. Multi-Agent Telemetry & Auditing (7 tables)

- **Database Scale**: Ingests 2,639 securities, 2,639 companies, and 2,715 exchange listings.

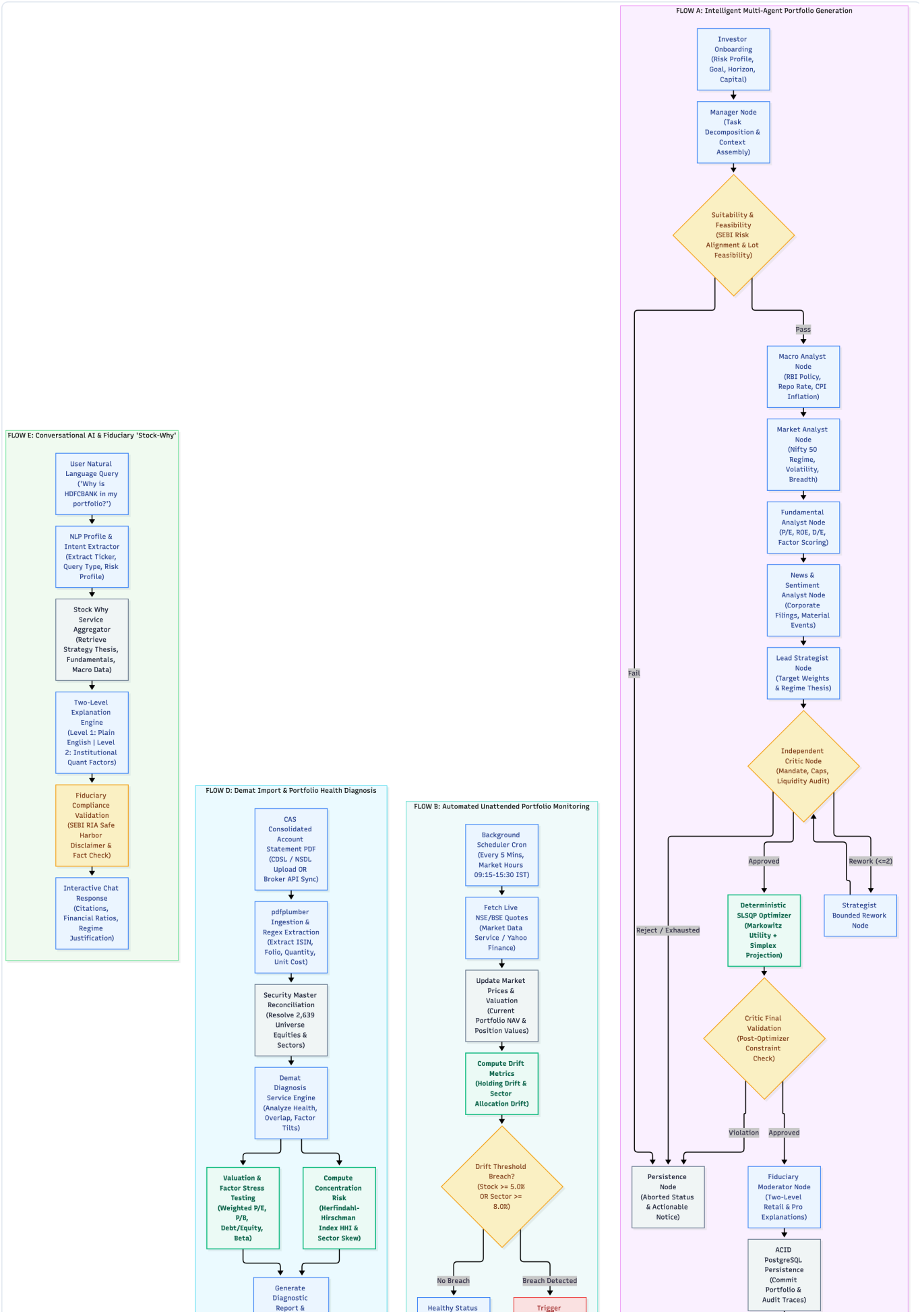
Layer 8: External Integrations & Market Data Layer

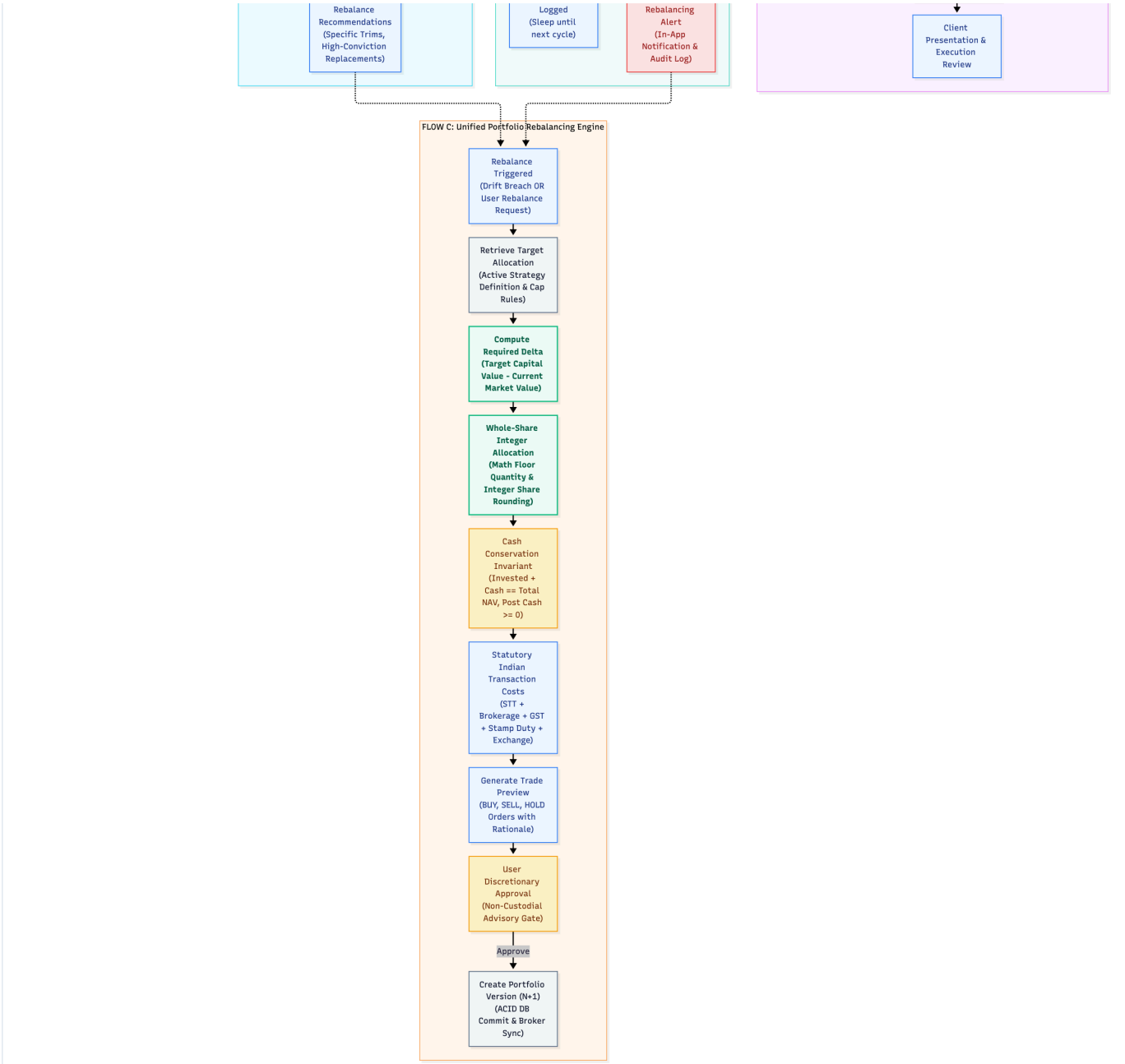
- **NSE India Archives**: Official master list of listed equities (`EQUITY_L.csv`).
- **BSE India Corporate Directory**: Universal ISIN matching and cross-exchange mapping.
- **Yahoo Finance (`yfinance`)**: Live market quotes, historical OHLCV data, and benchmark indices.
- **Reserve Bank of India (RBI)**: Macroeconomic policy rates and monetary indicators.
- **CDSL / NSDL**: Consolidated Account Statement (CAS) PDF import for existing demat portfolios.

Lucid — End-to-End Operational Methodology

This document details the complete mathematical, algorithmic, and agentic methodology governing Lucid's 5 core operational flows.

Operational Workflow Overview


FLOW A: Intelligent Multi-Agent Portfolio Generation
Investor Onboarding (Risk Profile, Goal, Horizon, Capital)
Manager Node (Task Decomposition & Context Assembly)
Suitability & Feasibility (SEBI Risk Alignment & Lot Feasibility)
Fail
Pass
Macro Analyst Node (RBI Policy, Repo Rate, CPI Inflation)
Market Analyst Node (Nifty 50 Regime, Volatility, Breadth)
Fundamental Analyst Node (P/E, ROE, D/E, Factor Scoring)
News & Sentiment Analyst Node (Corporate Filings, Material Events)
Lead Strategist Node (Target Weights & Regime Thesis)
Independent Critic Node (Mandate, Caps, Liquidity Audit)
Approved
Rework (<=2)
Reject / Exhausted
Deterministic SLSQP Optimizer (Markowitz Utility + Simplex Projection)
Strategist Bounded Rework Node
Critic Final Validation (Post-Optimizer Constraint Check)
Violation
Approved
Persistence Node (Aborted Status & Actionable Notice)
Fiduciary Moderator Node (Two-Level Retail & Pro Explanations)
ACID PostgreSQL Persistence (Commit Portfolio & Audit Traces)
FLOW B: Automated Unattended Portfolio Monitoring
Background Scheduler Cron (Every 5 Mins, Market Hours 09:15-15:30 IST)
Fetch Live NSE/BSE Quotes (Market Data Service / Yahoo Finance)
Update Market Prices & Valuation (Current Portfolio NAV & Position Values)
Compute Drift Metrics (Holding Drift & Sector Allocation Drift)
Drift Threshold Breach? (Stock >= 5.0% OR Sector >= 8.0%)
No Breach
Breach Detected
Healthy Status
Trigger
FLOW D: Demat Import & Portfolio Health Diagnosis
CAS Consolidated Account Statement PDF (CDSL / NSDL Upload OR Broker API Sync)
pdfplumber Ingestion & Regex Extraction (Extract ISIN, Folio, Quantity, Unit Cost)
Security Master Reconciliation (Resolve 2,639 Universe Equities & Sectors)
Demat Diagnosis Service Engine (Analyze Health, Overlap, Factor Tilts)
Valuation & Factor Stress Testing (Weighted P/E, P/B, Debt/Equity, Beta)
Compute Concentration Risk (Herfindahl-Hirschman Index HHI & Sector Skew)
Generate Diagnostic Report &
FLOW E: Conversational AI & Fiduciary 'Stock-Why'
User Natural Language Query ('Why is HDFCBANK in my portfolio?')
NLP Profile & Intent Extractor (Extract Ticker, Query Type, Risk Profile)
Stock Why Service Aggregator (Retrieve Strategy Thesis, Fundamentals, Macro Data)
Two-Level Explanation Engine (Level 1: Plain English | Level 2: Institutional Quant Factors)
Fiduciary Compliance Validation (SEBI RIA Safe Harbor Disclaimer & Fact Check)
Interactive Chat Response (Citations, Financial Ratios, Regime Justification)



FLOW A: Intelligent Multi-Agent Portfolio Generation Flow

Flow A translates an investor's subjective financial goals and risk tolerance into an optimized, mathematically verified equity portfolio.



1. Inputs

- **User Parameters**: Initial capital (e.g. ₹1,00,000), investment horizon (e.g. 5 years), risk tolerance profile (`conservative`, `moderate`, or `aggressive`), and goal type (`wealth_creation`, `retirement`, `capital_preservation`, `dividend_income`).
- **Database Universe**: 2,639 listed equities, filtered down to the 504 recommendation candidates passing liquidity criteria (daily turnover > ₹25,00,000).

2. Processing Steps

1. **Context Initialization**: `manager_node` parses goal parameters and establishes the typed `PortfolioAgentState`.

2. **Suitability & Feasibility**: `suitability_check_node` verifies that the requested capital can support integer shares without exceeding concentration caps (minimum viable capital is ₹10,000).
3. **Macroeconomic Analysis**: `macro_analyst_node` inspects RBI monetary policy data (repo rate, inflation) and sets broad equity asset allocation boundaries.
4. **Market Regime Classification**: `market_analyst_node` compares NIFTY 50 spot price against its 50-day and 200-day moving averages to determine market regime (Bullish, Neutral, Volatile).
5. **Fundamental Multi-Factor Ranking**: `fundamental_analyst_node` scores candidates across Quality, Value, Momentum, and Low-Volatility factors using sector-relative Winsorized Z-scores.
6. **Sentiment & Event Analysis**: `news_analyst_node` evaluates recent corporate announcements and flags material governance or earnings risks.
7. **Tactical Strategy Formulation**: `analyst_node` selects 10–18 target candidates and assigns preliminary baseline weights.
8. **Independent Fiduciary Audit**: `critic_node` enforces regulatory and policy limits:
    - Single stock weight $\le 10.0%$ (or $15.0%$ for aggressive).
    - Sector weight $\le 30.0%$ (or $35.0%$ for aggressive).
    - If violated, routes to `strategist_rework_node` (max 2 iterations).
9. **Deterministic Optimization**: `deterministic_optimizer_node` executes SciPy SLSQP to find mathematically optimal weights, followed by integer-share conversion via floor division.
10. **Post-Optimizer Final Validation**: `critic_final_validation_node` verifies that post-rounding integer weights remain strictly within policy bounds.
11. **Fiduciary Explanations**: `moderator_explanation_node` synthesizes dual-level rationales.
12. **ACID Persistence**: `persistence_node` commits the portfolio, version, decisions, and telemetry within a single database transaction.

3. Failure & Fallback Paths

- **Unsuitable/Infeasible Request**: If capital is below ₹10,000 or risk capacity is zero, the graph terminates early at `persistence_node` with status `ABORTED_INFEASIBLE` and returns an actionable explanation.
- **Critic Rejection Exhaustion**: If the strategist fails to satisfy constraints after 2 rework cycles, the workflow aborts safely without generating an invalid portfolio.
- **SLSQP Non-Convergence**: If the numerical solver fails, the system executes an iterative sector-constrained water-filling fallback algorithm.

FLOW B: Automated Unattended Portfolio Monitoring Flow

Flow B operates continuously in the background to ensure that active investor portfolios remain aligned with target allocations.



1. Inputs

- Active portfolio version ID and associated `portfolio_security_decisions`.
- Live NSE/BSE equity quotes retrieved from Yahoo Finance.

2. Processing Steps

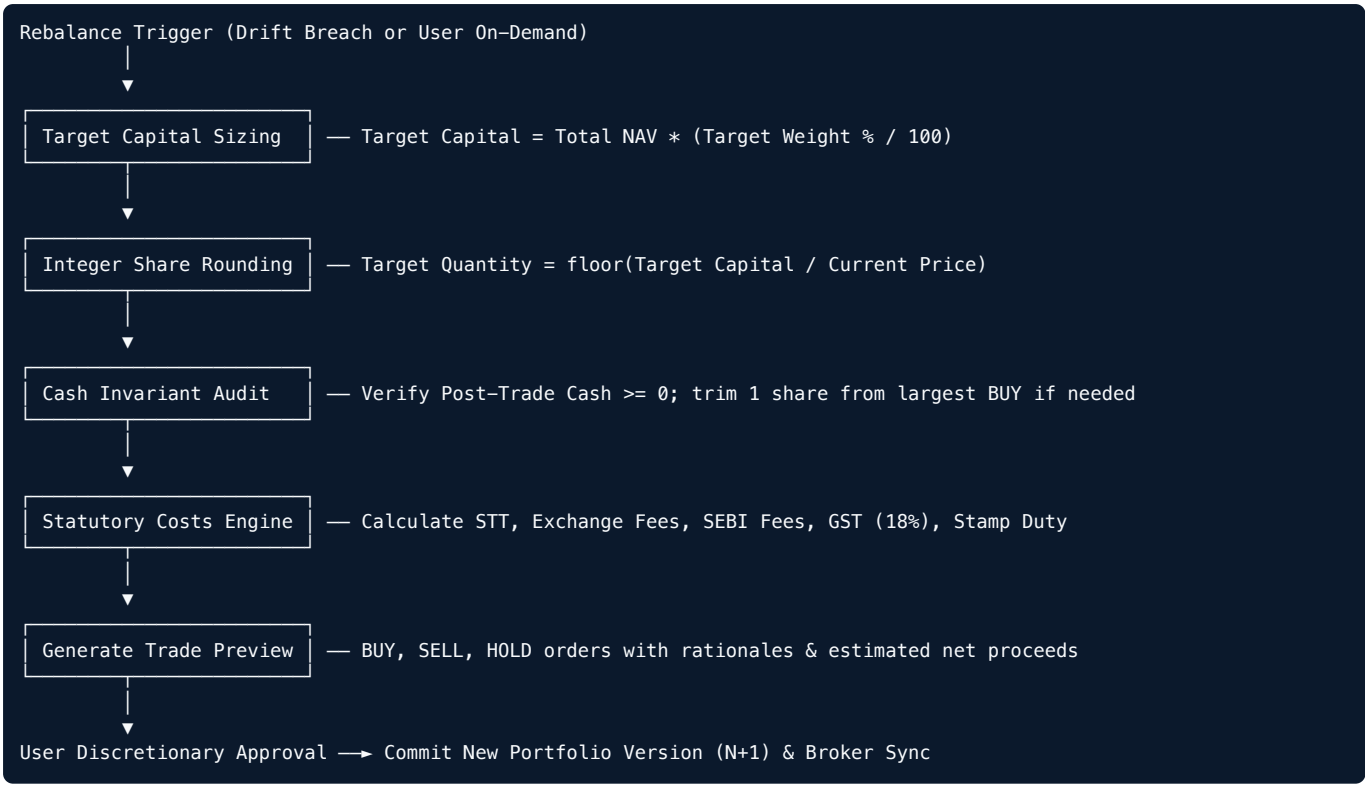
1. **Cron Trigger**: The `BackgroundMonitoringScheduler` wakes up every 300 seconds during active market hours (Monday–Friday, 09:15 – 15:30 IST).
2. **Batch Price Fetching**: Retrieves latest trading prices for all unique holdings across active portfolios.
3. **Valuation Update**: Computes: $$ ext{Current Position Value}_i = ext{Shares Quantity}_i imes ext{Latest Price}*i$$ $$ ext{Total Portfolio Value} = |sum{i=1}^n ext{Current Position Value}_i + ext{Unallocated Cash}$$ $$ ext{Current Weight}_i (%) = \left(rac{ ext{Current Position Value}_i}{ ext{Total Portfolio Value}} ight) imes 100$$
4. **Drift Computation**: $$ ext{Holding Drift}_i = ext{Current Weight}_i - ext{Target Weight}_i$$ $$ ext{Sector Drift}*k = |sum{i \in ext{Sector}_k} ext{Current Weight}*i - |sum{i \in ext{Sector}_k} ext{Target Weight}_i$$
5. **Breach Detection**: Flags a drift event if any holding $| ext{Holding Drift}_i| \ge 5.0%$ or any sector $| ext{Sector Drift}_k| \ge 8.0%$.

3. Outputs

- Health status (`HEALTHY` or `DRIFT_ALERT`).
- Dispatches transactional in-app notifications and creates audit events.

FLOW C: Unified Portfolio Rebalancing Engine

Flow C calculates the precise, whole-share buy and sell transactions needed to restore a drifted portfolio back to its target mandate.



1. Mathematical Invariants

- **Whole-Share Sizing**: No fractional shares; all trade quantities are discrete integers.
- **Cash Conservation**: Total invested capital plus residual cash must equal total portfolio NAV. Post-trade cash can never be negative. If integer rounding produces a minor deficit, the algorithm trims exactly 1 share from the largest BUY trade.
- **Statutory Indian Costs**: Automatically factors in:
  - STT: 0.10% on delivery buy and sell.
  - Exchange Transaction Fee: 0.00297% (NSE).
  - SEBI Turnover Fee: 0.0001%.
  - GST: 18% on (brokerage + exchange charges + SEBI charges).
  - Stamp Duty: 0.015% on delivery buy orders.

FLOW D: Demat Import & Health Diagnosis Flow

Flow D allows investors to upload their existing CDSL or NSDL Consolidated Account Statement (CAS) PDF and receive an immediate institutional health audit.

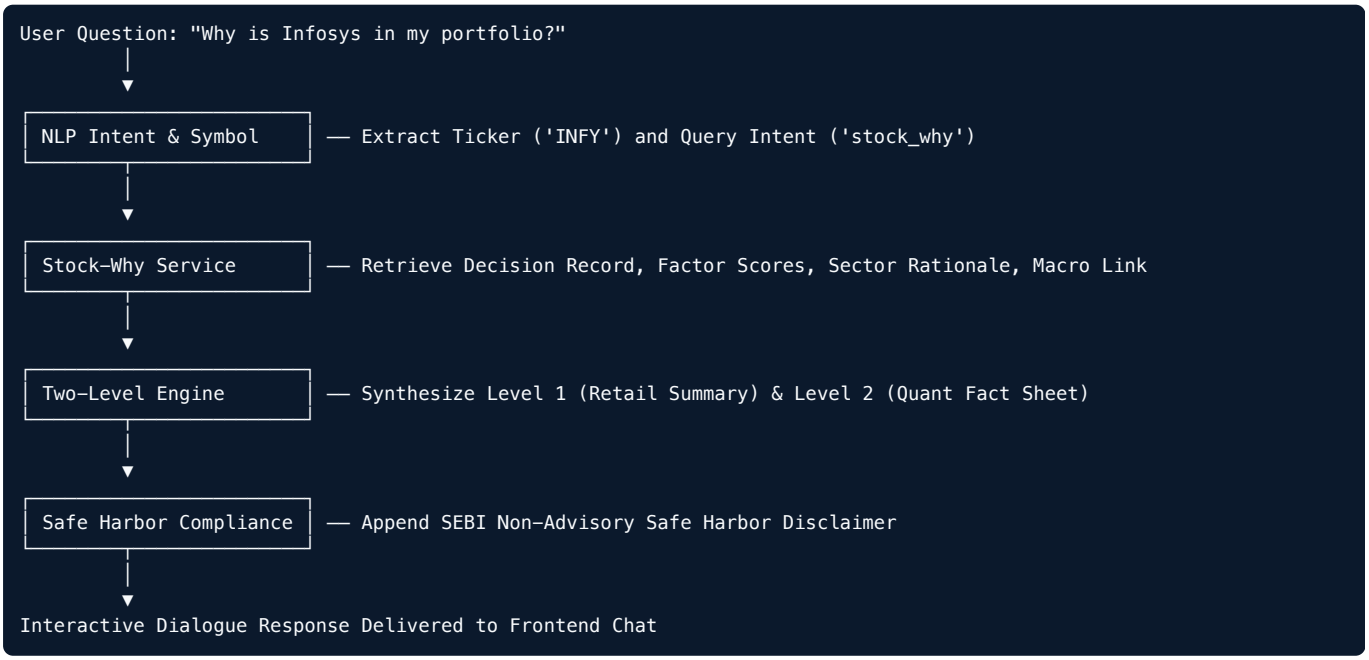


1. Key Metrics Evaluated

- **Herfindahl-Hirschman Index (HHI)**: $$ ext{HHI} = \sum_{i=1}^n (ext{Weight}_i%)^2$$
  - $ ext{HHI} < 1500$: Diversified.
  - $1500 \le ext{HHI} \le 2500$: Moderately Concentrated.
  - $ ext{HHI} > 2500$: Highly Concentrated (High Risk).
- **Sector Concentration**: Detects sectors with exposure exceeding 30%.
- **Penny Stock / Illiquidity Flags**: Highlights holdings outside the verified liquid universe.

FLOW E: Conversational AI & Fiduciary 'Stock-Why' Flow

Flow E provides plain-English, verifiable answers to investor questions regarding their portfolio allocations without hallucinating financial metrics.



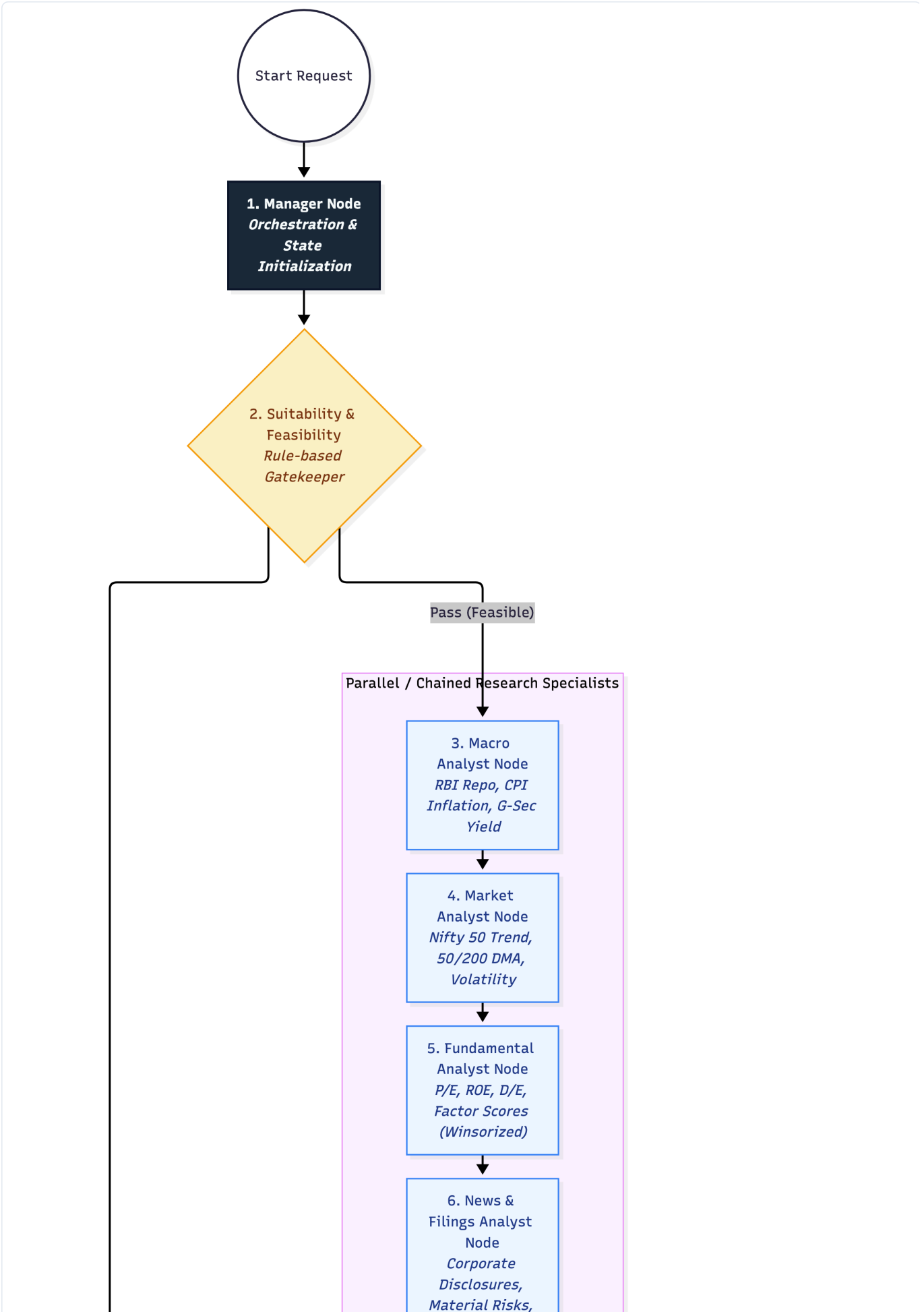
1. Dual-Level Explanation Model

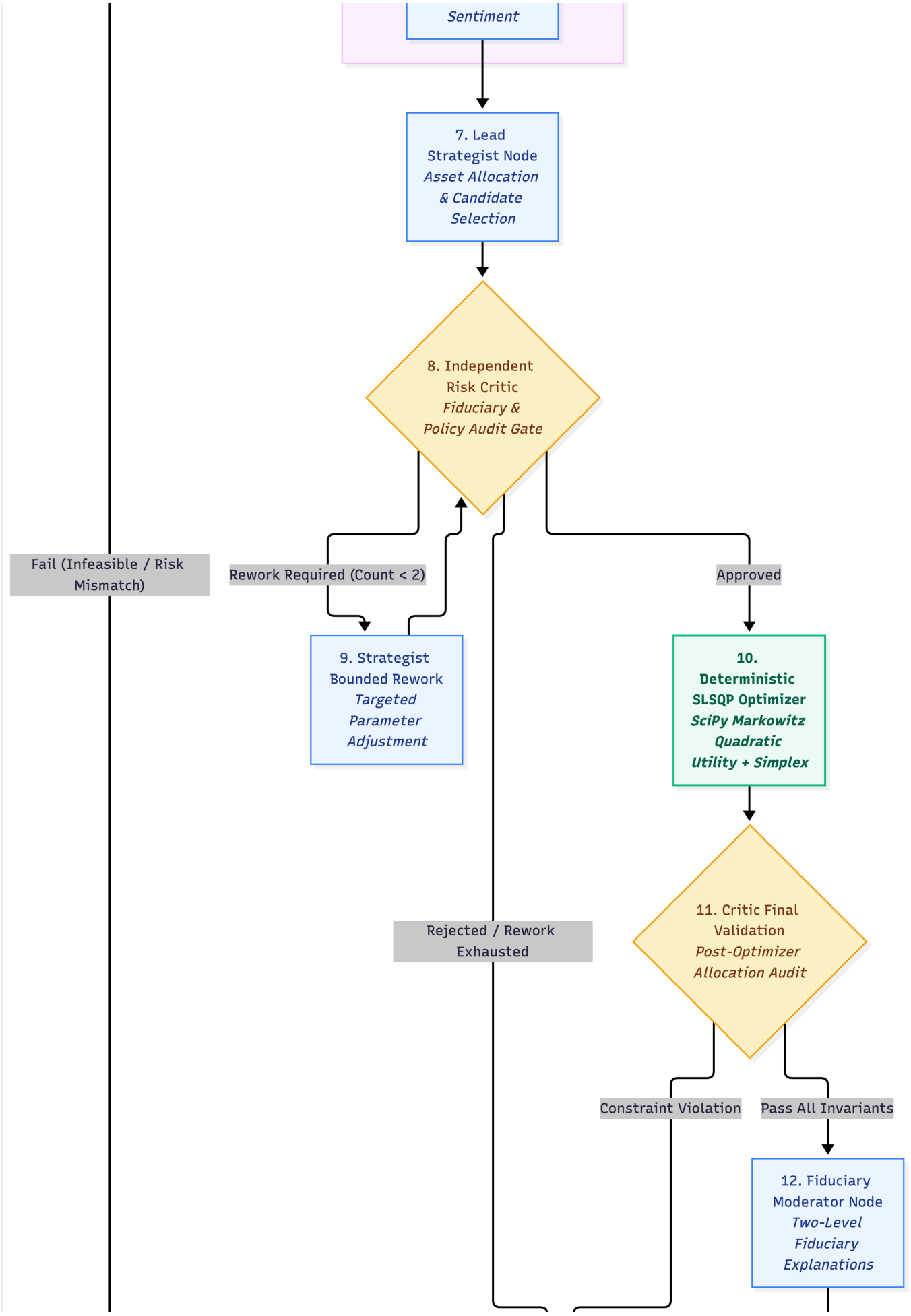
- **Level 1 (Executive Summary)**: Plain English explanation suitable for everyday retail investors (e.g. *"Infosys was included to provide defensive IT export exposure with consistent 28% return on equity and zero long-term debt"*).
- **Level 2 (Institutional Quantitative Fact Sheet)**: Verifiable metrics including exact factor Z-scores, historical beta, P/E ratio, 3-year CAGR, and macroeconomic regime alignment.

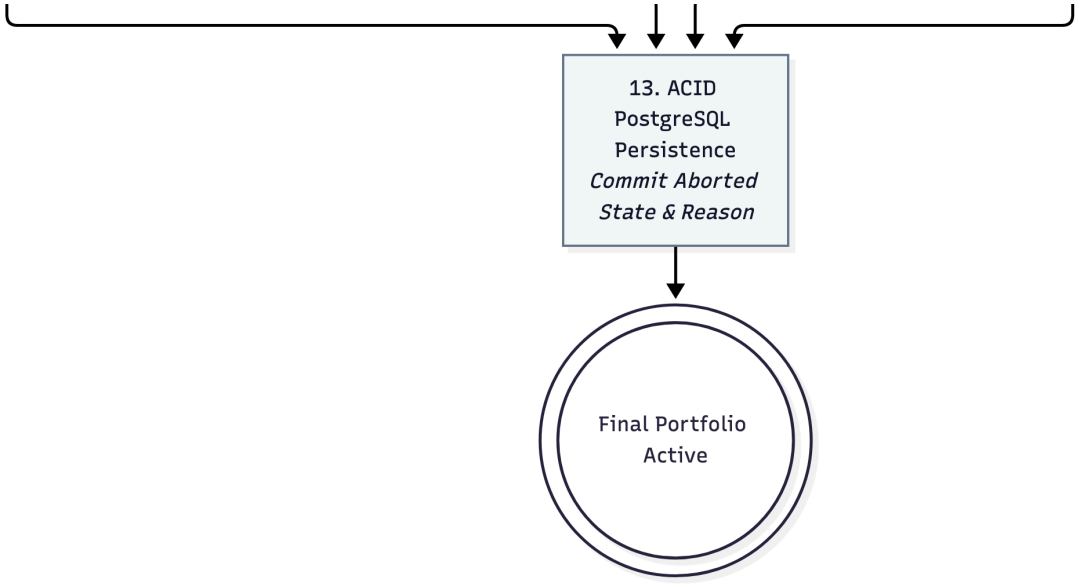
Lucid — Agentic AI Architecture & LangGraph Workflow

This document details the multi-agent artificial intelligence framework powering **Lucid**, implemented with LangGraph and executing across 13 specialized nodes.

Agentic AI Topology & State Transitions


Start Request
1. Manager Node
Orchestration & State Initialization
2. Suitability & Feasibility
Rule-based Gatekeeper
Pass (Feasible)
Parallel / Chained Research Specialists
3. Macro Analyst Node
RBI Repo, CPI Inflation, G-Sec Yield
4. Market Analyst Node
Nifty 50 Trend, 50/200 DMA, Volatility
5. Fundamental Analyst Node
P/E, ROE, D/E, Factor Scores (Winsorized)
6. News & Filings Analyst Node
Corporate Disclosures, Material Risks,


Sentiment
7. Lead Strategist Node Asset Allocation & Candidate Selection
8. Independent Risk Critic Fiduciary & Policy Audit Gate
Fail (Infeasible / Risk Mismatch)
Rework Required (Count < 2)
Approved
9. Strategist Bounded Rework Targeted Parameter Adjustment
10. Deterministic SLSQP Optimizer SciPy Markowitz Quadratic Utility + Simplex
Rejected / Rework Exhausted
11. Critic Final Validation Post-Optimizer Allocation Audit
Constraint Violation
Pass All Invariants
12. Fiduciary Moderator Node Two-Level Fiduciary Explanations




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graph TD
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    classDef dbNode fill:#F1F5F9,stroke:#64748B,stroke-width:1.5px,color:#0F172A;

    START((Start)) --> N1["1. Manager Node"]:::entryNode
    N1 --> N2{"2. Suitability & Feasibility"}:::ruleNode
    N2 -- "Fail" --> N13["13. ACID PostgreSQL Persistence"]:::dbNode
    N2 -- "Pass" --> N3["3. Macro Analyst Node"]:::llmNode
    N3 --> N4["4. Market Analyst Node"]:::llmNode
    N4 --> N5["5. Fundamental Analyst Node"]:::llmNode
    N5 --> N6["6. News & Filings Analyst Node"]:::llmNode
    N6 --> N7["7. Lead Strategist Node"]:::llmNode
    N7 --> N8{"8. Independent Risk Critic"}:::ruleNode
    N8 -- "Rework (<=2)" --> N9["9. Strategist Bounded Rework"]:::llmNode
    N9 --> N8
    N8 -- "Reject / Exhausted" --> N13
    N8 -- "Approved" --> N10["10. Deterministic SLSQP Optimizer"]:::mathNode
    N10 --> N11{"11. Critic Final Validation"}:::ruleNode
    N11 -- "Violation" --> N13
    N11 -- "Pass" --> N12["12. Fiduciary Moderator Node"]:::llmNode
    N12 --> N13
    N13 --> END_SUCCESS(((Final Active Portfolio)))
```

Detailed Node Specifications

#	Node Name	Execution Type	Primary Responsibilities	Input State Keys	Output State Keys
1	`manager_node`	Orchestration	Decomposes user request, parses financial goals and capital constraints, initializes state	`user_profile`, `goal`, `capital`, `risk_category`	`workflow_stage`, `target_return`, `active_constraints`
2	`suitability_check_node`	Rule-based	Validates investor risk capacity against regulatory bounds and verifies minimum capital viability (₹10,000)	`risk_category`, `capital`, `horizon_years`	`suitability_passed`, `suitability_notes`
3	`macro_analyst_node`	LLM + Tool	Analyzes RBI monetary policy, repo rate (6.50%), CPI inflation, and yield curve spreads	`macro_data`, `risk_category`	`macro_regime`, `equity_allocation_boundary`
4	`market_analyst_node`	LLM + Tool	Classifies NIFTY 50 trend, 50/200 DMA breadth, and market volatility regime	`nifty_historical_prices`, `india_vix`	`market_regime`, `market_trend_notes`
5	`fundamental_analyst_node`	LLM + Tool	Evaluates corporate quality (ROE > 15%), P/E valuation, debt leverage, and multi-factor Z-scores	`candidate_universe`, `fundamental_metrics`	`ranked_candidates`, `factor_scores`
6	`news_analyst_node`	LLM + Tool	Evaluates corporate disclosures, quarterly earnings announcements, and material sentiment risks	`ranked_candidates`, `market_news`	`news_risk_flags`, `candidate_sentiment`
7	`analyst_node` (Lead Strategist)	LLM	Synthesizes macroeconomic, market, and fundamental findings into a tactical portfolio selection (10–18 stocks)	`ranked_candidates`, `macro_regime`, `market_regime`	`selected_securities`, `preliminary_weights`
8	`critic_node` (Risk Critic)	Rule + LLM	Adversarially audits allocations against single-	`selected_securities`, `preliminary_weights`	`critic_verdict_status`, `critic_approved`, `critic_notes`

#	Node Name	Execution Type	Primary Responsibilities	Input State Keys	Output State Keys
			stock (10%) and sector (30%) concentration caps and minimum liquidity		
9	`strategist_rework_node`	LLM	Adjusts proposed candidates and weights based on Critic audit feedback (bounded to max 2 iterations)	`selected_securities`, `critic_notes`, `rework_count`	`selected_securities`, `rework_count`
10	`deterministic_optimizer_node`	Mathematical	Solves Markowitz quadratic utility via SciPy SLSQP; converts weights to integer shares via floor division	`selected_securities`, `active_constraints`, `capital`	`optimized_allocations`, `unallocated_cash`, `validation_passed`
11	`critic_final_validation_node`	Rule-based	Verifies post-optimizer integer share sizing and unallocated cash against policy caps	`optimized_allocations`, `unallocated_cash`	`critic_final_output`, `can_proceed`
12	`moderator_explanation_node`	LLM	Formulates dual-level fiduciary explanations: Level 1 Retail Summary and Level 2 Institutional Fact Sheet	`optimized_allocations`, `macro_regime`, `factor_scores`	`explanations`, `retail_summary`, `quant_factsheet`
13	`persistence_node`	Database	Commits portfolio, versions, individual security decisions, and audit events within an ACID PostgreSQL transaction	Complete `PortfolioAgentState`	`portfolio_id`, `version_id`, `status`

Fiduciary Safeguards & Anti-Hallucination Design

1. **Decoupled Mathematical Optimization**: The LLM never invents portfolio weights. Optimization is performed strictly by SciPy's numerical SLSQP solver in Python (`backend/portfolio_engine/optimizer.py`).
2. **Double-Critic Architecture**:
   - `critic_node` audits the strategic thesis *before* numerical optimization.
   - `critic_final_validation_node` audits the integer-lot allocations *after* numerical optimization.

3. **Hard Bounded Loops**: Rework loops between the Lead Strategist and Critic are strictly capped at `max_rework_count = 2`. If consensus is not reached, the system halts with a transparent rejection report.
4. **Mandatory Safe Harbor Notice**: Every explanation delivered to the investor includes statutory SEBI safe harbor disclosures, preventing the AI from presenting algorithmic forecasts as guaranteed financial returns.

Lucid — Database Architecture & Entity-Relationship Specifications

This document provides the authoritative structural and relational specification of Lucid's database, implemented in PostgreSQL 16 and managed via SQLAlchemy 2.0 ORM and Alembic migrations.

1. Master Relational Topology & Database Summary

Lucid's operational persistence layer consists of **62 relational tables** containing **565 structured attributes** connected by **102 foreign-key constraints** partitioned across **8 functional domains**:

Domain #	Functional Domain Name	Table Count	Attribute Count	Primary Purpose
Domain 1	User Identity, Authentication & Security	7	59	Manages user credentials, RBAC roles, audit sessions, tokens, and multi-factor auth
Domain 2	Investor Suitability, Profiling & Goals	5	47	Captures SEBI risk psychometrics, investor suitability tiers, and financial targets
Domain 3	Security Master & Corporate Taxonomy	9	56	Canonical asset directory (2,639 securities, 2,715 listings across NSE & BSE)
Domain 4	Market Data, Fundamentals & News	11	101	Stores OHLCV prices, multi-factor scores, economic indicators, and news sentiment
Domain 5	Portfolio Engine, Optimization & Explanations	13	133	Persists portfolio versions, SLSQP decisions, integer lots, and dual-level explanations
Domain 6	Broker Accounts & Order Execution	6	55	Tracks broker connections (Zerodha, Upstox), trade orders, and historical performance
Domain 7	Demat Statement Diagnosis & Rebalancing	4	38	Ingests CAS statements, computes HHI concentration, and tracks rebalance proposals
Domain 8	Multi-Agent Telemetry & System Auditing	7	76	Records step-by-step LangGraph node executions, LLM token costs, and notifications
TOTAL	**8 Operational Domains**	**62 Tables**	**565 Columns**	**100% Relational Integrity (PostgreSQL)**

2. Security Master Relational Model (Domain 3)

The Security Master resolves the fundamental real-world distinction between **Companies**, **Securities**, and **Exchange Listings**:

```
erDiagram
    COMPANIES ||--o{ SECURITIES : "issues (1:N)"
    SECURITIES ||--o{ INSTRUMENT_LISTINGS : "trades_on (1:N)"
    EXCHANGES ||--o{ INSTRUMENT_LISTINGS : "hosts (1:N)"
    SECTORS ||--o{ INDUSTRIES : "classifies (1:N)"
    INDUSTRIES ||--o{ COMPANIES : "categorizes (1:N)"

    COMPANIES {
        uuid company_id PK
        string legal_name
        string cin_number UK
        string pan_number
        uuid industry_id FK
    }

    SECURITIES {
        uuid security_id PK
        uuid company_id FK
        string isin UK
        string security_name
        string security_type
        string security_status
    }

    INSTRUMENT_LISTINGS {
        uuid listing_id PK
        uuid security_id FK
        uuid exchange_id FK
        string symbol
        string series
        string trading_status
    }

    EXCHANGES {
        uuid exchange_id PK
        string exchange_code UK
        string exchange_name
    }
```

Relational Hierarchy:

1. **`Company`**: Represents the underlying legal entity (e.g. *Tata Consultancy Services Limited*, CIN: `L22210MH1995PLC084781`).
2. **`Security`**: Represents the distinct financial asset issued by that company, globally identified by its unique International Securities Identification Number (ISIN, e.g. `INE467B01029`).
3. **`InstrumentListing`**: Represents where and how that security is actively traded on a specific exchange:
    - On the **National Stock Exchange (NSE)**, it trades under ticker `TCS` with series `EQ`.
    - On the **Bombay Stock Exchange (BSE)**, it trades under scrip code `532540` / ticker `TCS`.

Ingestion Scale Verification:

- **Total Registered Companies**: 2,639
- **Total Securities (Unique ISINs)**: 2,639 (100% unique)
- **Total Listings**: 2,715
    - NSE Listings: 2,640 (Series EQ, BE, BZ)
    - BSE Verified Listings: 72
    - Synthetic Demo Listings: 3
- **Active Tradeable Listings**: 2,606 (Series EQ and BE on NSE)
- **Suspended Listings**: 34 (Series BZ on NSE, retained for portfolio diagnosis of legacy demat accounts)

---

3. Detailed Specifications of the 8 Domains

Domain 1: User Identity & Authentication (7 Tables)

- **`app_users`**: Core user accounts, email, hashed password, role (`investor`, `admin`, `advisor`), status.

- `user_profiles`: Investor KYC metadata, phone number, PAN status, onboarding completion timestamp.
- `user_preferences`: Currency display preference, theme, notification channels.
- `user_consents`: Fiduciary safe harbor acknowledgement, terms of service consent, SEBI disclaimer signatures.
- `user_verification_codes`: Temporary OTP codes for multi-factor authentication and email confirmation.
- `revoked_tokens`: Blacklisted JWT token identifiers (JTI) for instantaneous session invalidation.
- `user_profile_versions`: Audit log capturing historical modifications to user KYC parameters.

Domain 2: Investor Suitability & Goals (5 Tables)

- `financial_goals`: Individual targets (e.g., "Retirement Fund", "Down Payment"), target capital, horizon months, priority.
- `risk_questions`: Standardized psychometric questionnaire items evaluating loss tolerance and investment experience.
- `risk_question_options`: Weighted answer choices contributing to the numerical risk score.
- `risk_assessments`: Evaluated assessment sessions mapping raw scores to risk categories (`conservative`, `moderate`, `aggressive`).
- `risk_assessment_answers`: Specific answers submitted by the investor, preserving evidentiary audit trails.

Domain 3: Security Master & Corporate Taxonomy (9 Tables)

- `companies`: Legal corporate entities, incorporation date, CIN, headquarter location.
- `exchanges`: Registered stock exchanges (NSE, BSE, DEMOEX).
- `sectors`: Macro sectors (e.g., Financials, Technology, Industrials, Healthcare).
- `industries`: Granular industries linked to parent sectors.
- `company_industries`: Cross-reference association mapping multi-segment conglomerates.
- `securities`: Unique asset definitions tied to ISINs and currency codes.
- `instrument_listings`: Specific exchange scrip symbols, tick sizes, lot sizes, and trading series.
- `user_sector_preferences`: Investor-configured sector whitelists or exclusion overrides.
- `user_security_preferences`: Investor-configured stock exclusions (e.g. tobacco, defense).

Domain 4: Market Data, Prices & Fundamentals (11 Tables)

- `data_sources`: Ingestion registry tracking external providers (NSE, BSE, Yahoo Finance, RBI).
- `data_ingestion_runs`: Ingestion batch records tracking status, start/end timestamps, and record counts.
- `data_ingestion_errors`: Detailed ingestion failure logs storing raw payloads and HTTP status codes.
- `source_instrument_mappings`: Translation table bridging exchange tickers to vendor tickers (e.g., `RELIANCE` $ o$ `RELIANCE.NS`).
- `market_prices`: Historical and latest OHLCV price observations and trading volumes.
- `metric_definitions`: Canonical definitions of fundamental indicators (P/E, ROE, ROCE, D/E).
- `fundamental_metrics`: Normalized financial ratios and Winsorized factor scores.
- `economic_indicators`: Macroeconomic indicators (RBI Repo Rate, CPI Inflation, G-Sec Yield).
- `economic_observations`: Time-series observation values for macroeconomic indicators.
- `market_news_articles`: Market headlines, corporate disclosure summaries, and sentiment scores.
- `news_article_securities`: Many-to-many association mapping news events to impacted securities.

Domain 5: Portfolio Engine, Optimization & Explanations (13 Tables)

- `investor_portfolios`: Top-level portfolio containers linking investors to investment goals.
- `portfolio_generation_requests`: Asynchronous generation requests tracking multi-agent execution state.
- `strategy_definitions`: Mathematical allocation strategy rules, risk parameters, and asset bounds.
- `strategy_versions`: Versioned snapshots of investment strategies over time.
- `portfolio_versions`: Versioned portfolio states (v1, v2, v3...) tracking revisions and rebalance bumps.
- `portfolio_security_decisions`: Specific holding allocations storing target weight %, integer shares, and actual invested amounts.
- `portfolio_version_constraints`: Snapshot of hard policy constraints enforced during optimization.
- `evidence_items`: Atomic financial facts (e.g. ROE=24.5%, Beta=0.85) cited during generation.

- `recommendation_explanations`: Dual-level text explanations (Retail Plain English + Institutional Quant Fact Sheet).
- `explanation_evidences`: Association table linking explanations to underlying evidence items.
- `portfolio_validation_runs`: Pre- and post-optimization constraint audit execution logs.
- `portfolio_validation_results`: Individual constraint test outcomes (e.g. Single-Stock-Cap: PASS).
- `portfolio_scenarios`: Stress test simulations (e.g. 2008 Crash, 2020 COVID shock, Rate Spike).

Domain 6: Broker Accounts & Execution (6 Tables)

- `broker_providers`: Registry of supported broker APIs (Zerodha Kite Connect, Upstox, Groww).
- `user_broker_accounts`: Encrypted broker credentials, OAuth refresh tokens, and account status.
- `broker_orders`: Individual trade orders generated during initial investment or rebalancing.
- `broker_order_events`: Audit lifecycle events (SUBMITTED, ACKNOWLEDGED, FILLED, REJECTED).
- `broker_executions`: Execution fill records storing exact fill prices, quantities, and execution exchange.
- `portfolio_performance_snapshots`: Periodic portfolio valuation, NAV, and return records.

Domain 7: Demat Diagnosis & Rebalancing (4 Tables)

- `existing_portfolios`: Uploaded CAS Demat portfolios imported via CDSL/NSDL PDF statements.
- `existing_portfolio_holdings`: Individual parsed holding lines (ISIN, quantity, acquisition cost, market value).
- `portfolio_diagnoses`: Comprehensive health assessment results (HHI concentration score, sector overlap, risk flags).
- `rebalance_suggestions`: Specific suggested whole-share BUY/SELL trades to remediate health flags.

Domain 8: Multi-Agent Telemetry & Auditing (7 Tables)

- `agent_definitions`: Registry of the 13 specialized agent nodes and their operational personas.
- `agent_versions`: Version control for agent prompts, toolsets, and system instructions.
- `agent_execution_runs`: Granular per-node execution telemetry (latency, tokens consumed, prompt, response).
- `knowledge_documents`: Vector-indexed fiduciary rules, SEBI guidelines, and financial research documents.
- `user_feedbacks`: Explicit investor feedback on generated portfolios and explanations.
- `notifications`: Transactional in-app notifications (Drift alerts, rebalance previews, security notices).
- `audit_events`: Immutable system audit trail capturing every state change, trade approval, and policy override.

Lucid — Mathematical & Portfolio Optimization Methodology

This document details the rigorous financial formulas, mathematical models, optimization algorithms, and execution invariants implemented across Lucid's portfolio engine.

1. Portfolio Optimization Formulation (Markowitz SLSQP)

Lucid executes deterministic mean-variance optimization locally in Python using `scipy.optimize.minimize` with the **Sequential Least Squares Programming (SLSQP)** algorithm (`backend/portfolio_engine/optimizer.py`).

Mathematical Objective Function

The optimizer minimizes the negative risk-adjusted quadratic utility:

$$\min_{w} \quad \mathcal{U}(w) = \frac{1}{2} \lambda \cdot w^T \Sigma w - \mu^T w$$

Where:

- $w = [w_1, w_2, \dots, w_n]^T$: Vector of candidate allocation weights.
- λ: Investor-specific risk aversion parameter derived from psychometric profiling:
  - **Conservative**: $\lambda = 5.0$
  - **Moderate**: $\lambda = 3.5$
  - **Aggressive**: $\lambda = 2.0$
- $\Sigma \in \mathbb{R}^{n \times n}$: Sample covariance matrix computed from annualized asset return volatilities and pairwise sector correlations: $$\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$$ where $\rho_{ii} = 1.0$, intra-sector correlation $\rho_{ij} = 0.50$, and cross-sector correlation $\rho_{ij} = 0.20$. A small Tikhonov regularization term ($\epsilon = 10^{-6} \mathbf{I}$) is added along the diagonal to ensure positive definiteness.
- $\mu = [E[R_1], E[R_2], \dots, E[R_n]]^T$: Vector of expected annualized asset returns.

Hard Institutional Constraints

1. **Full Investment (Simplex Budget Equality)**: $$\sum_{i=1}^n w_i = 1.0 \quad (100.0\%)$$
2. **Long-Only Asset Bounds (No Short Selling)**: $$0.0 \le w_i \le w_{\max} \quad \forall i \in {1, \dots, n}$$ where $w_{\max} = 10.0\%$ for moderate portfolios (15.0% for aggressive).
3. **Sector Concentration Inequality Constraints**: $$\sum_{i \in \text{Sector}*k} w_i |le S*{\max} \quad \forall k \in {1, \dots, K}$$ where $S_{\max} = 30.0\%$ for moderate portfolios (35.0% for aggressive).

Deterministic Constrained Fallback Solver

If the SLSQP numerical solver fails to converge due to tight boundary conditions, Lucid executes an **Iterative Sector-Constrained Water-Filling Algorithm** (`_sector_constrained_fallback`):

- Checks feasibility: Verifies that the candidate universe contains at least $\lceil 1 / S_{\max} \rceil$ distinct sectors.
- Iteratively allocates incremental weights $\Delta w = \min(\text{remaining} / m, w_{\max} - w_i, S_{\max} - S_k)$ across eligible assets until the simplex sum strictly equals 1.0.
- Returns explicit mathematical infeasibility if constraints cannot be satisfied without violating caps.

2. Integer Share Allocation & Cash Conservation Invariant

Real-world Indian retail equities cannot be purchased in fractional shares. Lucid translates continuous decimal weights into discrete integer lots while preserving total capital (`backend/portfolio_engine/allocation_rounding.py`).

Integer Lot Formula

$$\text{Target Capital}_i = \text{Total Investment Amount} \times \left(\frac{w_i}{100.0}\right)$$ $$\text{Shares Quantity}_i = \left\lfloor \frac{\text{Target Capital}_i}{\text{Latest Price}_i} \right\rfloor$$ $$\text{Actual Invested Amount}_i = \text{Shares Quantity}_i \times \text{Latest Price}_i$$

Strict Cash Invariant

$$\text{Total Invested} = \sum_{i=1}^n \text{Actual Invested Amount}_i$$ $$\text{Unallocated Cash} = \text{Total Investment Amount} - \text{Total Invested}$$

Mathematical Invariant: $$\text{Total Invested} + \text{Unallocated Cash} \equiv \text{Total Investment Amount}$$ $$\text{Unallocated Cash} \ge 0.00 \quad \text{(No negative cash balance / no uncollateralized margin)}$$

---

3. Financial Metrics & Econometric Formulas

Implemented in `backend/portfolio_engine/math_engine.py`:

3.1 Earnings Yield (E/P)

Handles operating losses and extreme valuations gracefully: $$\text{Earnings Yield} = \begin{cases} \min\left(\frac{1.0}{\text{P/E}}, 0.50\right) & \text{if } \text{P/E} > 0 \[6pt] \max\left(\frac{1.0}{\text{P/E}}, -0.25\right) & \text{if } \text{P/E} < 0 \text{ (Operating Losses)} \[6pt] \text{None} & \text{if } \text{P/E} = 0 \text{ or missing} \end{cases}$$

3.2 Compounded Annual Growth Rate (CAGR)

Computed using exact day-count fractions based on Indian trading days (252 days/year): $$\text{CAGR} = \left(\frac{\text{Price}*{T}}{|text{Price}*{0}}\right)^{\frac{252}{D}} - 1.0$$ where D is the number of actual trading days observed ($D \ge 30$).

3.3 Systematic Beta & Total Annualized Volatility

Treated as distinct risk dimensions: $$\text{Total Volatility } (\sigma_i) = \text{Std}(R_i) \times \sqrt{252}$$ $$\text{Systematic Beta } (\beta_i) = \frac{\text{Cov}(R_i, R_{\text{benchmark}})}{\text{Var}(R_{\text{benchmark}})}$$ Requires at least 60 daily trading observations.

3.4 Winsorized Sector-Relative Z-Scores

Protects factor models against extreme corporate outliers:

1. Clip metric values at the 5th percentile ($q_{0.05}$) and 95th percentile ($q_{0.95}$): $$\tilde{x}*i = |text{clip}(x_i, q*{0.05}, q_{0.95})$$
2. Compute sector mean μ_{sec} and standard deviation σ_{sec}.
3. Compute clamped Z-score: $$Z_i = \text{clip}\left(\frac{\tilde{x}*i - |mu*{\text{sec}}}{\sigma_{\text{sec}}}, -3.0, +3.0\right)$$

3.5 Expected Annual Return Formula

Strictly distinguished from unitless factor Z-scores; blends CAPM systematic risk premium with multi-factor alpha tilts:

$$E[R_i] = R_{\text{base}} + 1.20 \cdot (w_q Z_{q,i} + w_v Z_{v,i} + w_m Z_{m,i}) + 1.50 \cdot (\beta_i - 1.0)$$

Where:

- R_{base}: Base return target (e.g. $12.5%$ for moderate).
- w_q, w_v, w_m: Goal-specific factor loadings (Quality, Value, Momentum) summing to 1.0.
- Result is bounded within the institutional corridor $[4.0%, 30.0%]$.

---

4. Portfolio Monitoring & Drift Formulas

Implemented in `backend/portfolio_engine/monitoring_service.py` and `rebalancing_service.py`:

Asset Allocation Drift

$$\text{Holding Drift}_i = \text{Current Weight}_i - \text{Target Weight}_i$$ Trigger condition: $|\text{Holding Drift}_i| \ge 5.0%$.

Sector Allocation Drift

$$\text{Sector Drift}*k = |sum*{i \in \text{Sector}_k} \text{Current Weight}*i - |sum*{i \in \text{Sector}_k} \text{Target Weight}_i$$ Trigger condition: $|\text{Sector Drift}_k| \ge 8.0%$.

Rebalancing Turnover Metric

$$\text{Turnover} = \frac{1}{2} \sum_{i=1}^n |w_{\text{post}, i} - w_{\text{current}, i}|$$

5. Indian Statutory Equity Transaction Costs

Implemented in `backend/portfolio_engine/transaction_costs.py`:

Fee Component	Statutory Rate	Tax Base
Securities Transaction Tax (STT)	0.10%	Total delivery trade turnover (BUY and SELL)
Exchange Transaction Charges (NSE)	0.00297%	Total trade turnover
SEBI Turnover Fee	0.0001% (₹10 per crore)	Total trade turnover
Goods & Services Tax (GST)	18.0%	On (Brokerage + Exchange Fees + SEBI Fees)
Stamp Duty	0.015% (₹1500 per crore)	On delivery BUY trade value only
Brokerage	Flat ₹20 or 0.03%	Per executed trade order

Lucid — Security Architecture & Production Reliability

This document outlines the security controls, authentication mechanisms, network hardening, rate limiting, and failure containment architectures protecting the Lucid platform.

---

1. Authentication & Session Security

Implemented across `backend/auth/` and `backend/api/v1/auth.py`:

1.1 Password Hashing & Key Derivation

- Passwords are never stored in plaintext. Lucid utilizes **bcrypt** via Python `passlib` with work factor salt rounds ($2^{12} = 4096$ iterations).
- Immune to rainbow table lookups and brute-force GPU attacks.

1.2 JWT Token Lifecycle & Instant Session Invalidation

- **Access Tokens**: Short-lived (60 minutes) JSON Web Tokens signed using HMAC-SHA256 (`HS256`).
- **Refresh Tokens**: Cryptographically random UUID tokens stored securely to enable seamless session renewal without re-entering credentials.
- **Revoked Token Blacklist (`revoked_tokens` table)**: Upon user logout or security breach detection, token JTIs (JWT IDs) are persisted in PostgreSQL. Incoming requests with blacklisted tokens are immediately rejected with HTTP 401 Unauthorized.

1.3 Role-Based Access Control (RBAC)

- Strongly enforced user roles:
  - `investor`: Standard retail permissions (view own portfolios, run diagnosis, generate recommendations).
  - `advisor`: Professional permissions (review client allocations, audit suitability).
  - `admin`: Operational permissions (trigger full exchange synchronizations, inspect system telemetry).

---

2. HTTP Security Hardening & Gateway Protection

Implemented in `backend/main.py` via custom Starlette and FastAPI middleware:

2.1 Enterprise HTTP Security Headers

Every HTTP response generated by FastAPI is enriched with defensive headers:

- `X-Content-Type-Options: nosniff`: Prevents MIME-type sniffing.
- `X-Frame-Options: DENY`: Defends against clickjacking in iframes.
- `X-XSS-Protection: 1; mode=block`: Activates browser cross-site scripting filters.
- `Referrer-Policy: strict-origin-when-cross-origin`: Restricts referrer leakage.
- `Permissions-Policy: geolocation=(), camera=(), microphone=()`: Blocks hardware API access.
- `Strict-Transport-Security: max-age=31536000; includeSubDomains`: Mandates HTTPS in staging and production environments.

2.2 Strict CORS Policy

Restricts cross-origin browser requests strictly to authorized frontend origins (e.g., `http://localhost:3000`). Wildcard `*` origins are prohibited.

---

3. Rate Limiting & Denial-of-Service Defense

Implemented via `InMemoryRateLimiterMiddleware` (`backend/middleware/rate_limit.py`):

- **Sliding-Window In-Memory Tracking**: Tracks IP addresses and API keys with millisecond accuracy.

- **Tiered Endpoint Protection**:
  - `Authentication Endpoints` (`/api/v1/auth/login`, `/register`): Capped at 10 requests per minute to prevent credential stuffing.
  - `Agentic AI Generation` (`/api/v1/portfolios/generate`): Capped at 5 requests per minute to prevent compute exhaustion and protect LLM token quotas.
  - `Standard Market Data Endpoints`: Capped at 60 requests per minute.
- Returns standard HTTP 429 Too Many Requests with `Retry-After` headers when limits are exceeded.

---

4. Failsafe Architecture & Circuit Breakers

4.1 Production Security Key Validation (Fails Closed)

Implemented in `backend/config.py`:

```
def validate_security_keys(self) -> None:
    if self.app_env.lower() in ("staging", "production"):
        if not self.jwt_secret_key or "change-me" in self.jwt_secret_key or len(self.jwt_secret_key) < 32:
            raise ValueError(
                f"CRITICAL SECURITY CONFIGURATION ERROR: Environment '{self.app_env}' strictly requires "
                f"a non-default JWT_SECRET_KEY of at least 32 characters (RFC 7518 Section 3.2)."
            )
        if not self.app_secret_key or "change-me" in self.app_secret_key:
            raise ValueError(
                f"CRITICAL SECURITY CONFIGURATION ERROR: Environment '{self.app_env}' strictly requires "
                f"a non-default APP_SECRET_KEY."
            )
```

The application intentionally halts at startup if default, insecure, or truncated keys are configured in production.

4.2 External API Resilience

- **BSE Anti-Bot Defense**: When `api.bseindia.com` blocks automated headless requests with HTTP 403 Forbidden (Cloudflare Turnstile), Lucid never attempts to bypass or hack the gate. Instead, it seamlessly resolves BSE equity listings via SEBI-mandated universal ISIN matching and local verified registries.
- **Yahoo Finance Fallback**: Caches historical price data in PostgreSQL `market_prices`. If external endpoints throttle or experience network failure, the platform operates transparently on latest cached closing prices.
- **RBI Macroeconomic Fallback**: Built-in institutional baseline defaults (Repo Rate: 6.50%, CPI: 5.1%) ensure that monetary policy analysis never stalls during central bank portal updates.

4.3 Non-Custodial Advisory Boundary

- Lucid is strictly an **intelligent advisory platform**, not an automated execution bot.
- All broker orders (BUY/SELL) generated during initial allocation or rebalancing require explicit, discretionary investor approval in the UI before transmission. Lucid never triggers automated broker executions without user consent.